

 poolside/**Laguna-XS-2.1-INT4** 

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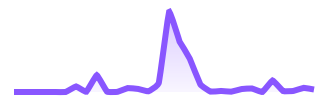
Laguna XS 2.1-INT4

Laguna XS 2.1-INT4 is a 33B total parameter Mixture-of-Experts model with 3B activated parameters per token designed for agentic coding and long-horizon work on a local machine. It uses Sliding Window Attention with per-head gating in 30 out of 40 layers for fast inference and low KV cache requirements.

This is the INT4 variant with an FP8-quantized KV cache. The [BF16](#), [FP8](#) and [NVFP4](#) variants are also available on Hugging Face.

Highlights

Downloads last month
6,629



 **Safetensors** 

Model size 41B params

Tensor type F32 · I32 · BF16

 Chat template

 Files info

 **Inference Providers** **NEW**

 Text Generation

This model isn't deployed by any Inference Provider.

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 **Model tree for poolside/Laguna-X...**

Base model poolside/Laguna-XS-2.1

 Quantized (22) [this model](#)

Finetunes [1 model](#)

 **Collection including poolside/Lag...**

Laguna XS 2.1  Collection

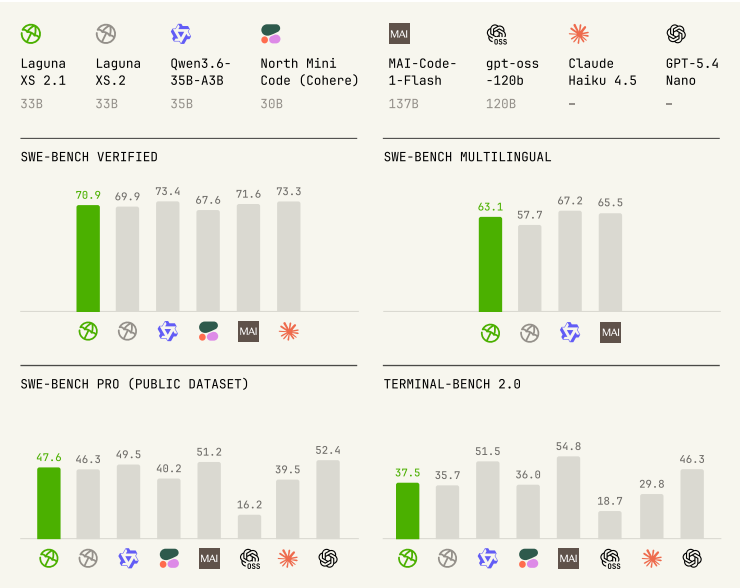
- **Mixed SWA and global attention layout:**
Laguna XS 2.1 uses sigmoid gating with per-layer rotary scales, enabling mixed SWA (Sliding Window Attention) and global attention layers in a 3:1 ratio (across 40 total layers)
 - **KV cache in FP8:** KV cache quantized to FP8, reducing memory per token
 - **Native reasoning support:** Interleaved thinking between tool calls with support for enabling and disabling thinking per-request
 - **Local-ready:** At 33B total parameters and 3B activated, Laguna XS 2.1 is compact enough to run on a Mac with 36 GB of RAM. Available on [Ollama](#) and [llama.cpp](#) (BF16 and Q4_K_M only)
 - **OpenMDW-1.1 license:** Use and modify the model and associated materials freely for commercial and non-commercial purposes ([learn more about OpenMDW](#))
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Model overview

- Training: pre-training, post-training and reinforcement learning stages
- Number of parameters: 33B total with 3B activated per token
- Optimizer: Muon
- Layers: 40 layers (10 layers with global attention, 30 layers with sliding window attention)
- Experts: 256 experts with 1 shared expert

- Sliding Window: 512 tokens
- Modality: text-to-text
- Context window: 262,144 tokens
- Reasoning support: interleaved thinking with preserved thinking

Benchmark results



Model	Size (total params.)	SWE-bench Verified	SWE-bench Multilingual	SWE-Bench Pro (Public Dataset)
Laguna XS 2.1	33B	70.9%	63.1%	47.6%
Laguna XS.2	33B	69.9%	57.7%	46.3%
Qwen3.6-35B-A3B	35B	73.4%	67.2%	49.5%

Model	Size (total params.)	SWE- bench Verified	SWE-bench Multilingual	SWE- Bench Pro (Public Dataset)
North Mini Code	30B	67.6%	-	40.2%
MAI- Code-1- Flash	137B	71.6%	65.5%	51.2%
gpt-oss- 120B	120B	-	-	16.2%
Claude Haiku 4.5	-	73.3%	-	39.5%
GPT-5.4 Nano	-	-	-	52.4%

We used the highest publicly-referenced scores for all comparison models across each benchmark. In all cases these were official scores published in release blog posts or equivalent, with the exception of gpt-oss-120b and Claude Haiku 4.5 where the highest published (verified) scores for SWE-Bench Pro and Terminal-Bench 2.0 are from their respective official leaderboards.

► Expand for benchmarking methodology

Quantization quality

Laguna XS 2.1-INT4 was evaluated against the unquantized BF16 checkpoint on the same four agentic benchmarks. Scores are mean pass@1; the ±

figure is the run-to-run variation and *RH* is the share of runs flagged by our post-hoc reward-hack judge (see the benchmarking methodology above).

Benchmark	Laguna XS 2.1 (BF16)	Laguna XS 2.1-INT4
SWE-bench Verified	70.85% ± 0.85% (6.51% RH)	71.70% ± 0.88% (5.91% RH)
SWE-bench Multilingual	63.17% ± 1.42% (7.25% RH)	63.42% ± 1.37% (7.45% RH)
SWE-Bench Pro (Public Dataset)	47.61% ± 0.96% (2.40% RH)	48.70% ± 0.89% (3.29% RH)
Terminal-Bench 2.0	37.53% ± 2.81% (2.12% RH)	37.08% ± 2.70% (3.04% RH)

Usage

Laguna XS 2.1-INT4 has launch-day support in vLLM and Transformers.

For complete usage instructions, see the main [Laguna XS 2.1 model card](#).

Local deployment

Laguna XS 2.1-INT4 is supported in vLLM and Transformers. Use Laguna XS 2.1 with Ollama (with MLX support) or Llama.cpp (BF16 and Q4_K_M only) for the best results on your local machine.

vLLM

The full vLLM recipe is on the main [Laguna XS 2.1 model card](#) and on the [vLLM recipes page](#).

Quantization is detected automatically from `quantization_config` in this checkpoint, so the same command works with `poolside/Laguna-XS-2.1-INT4` substituted for the model ID. No extra flags required.

INT4 support landed in vLLM with [vllm-project/vllm#47154](#). This checkpoint mixes INT4 and group-quantized INT8 experts, which earlier builds rejected in the Marlin WNA16 MoE path. Use a vLLM build that includes this fix.

Tool calling requires a vLLM build with [vllm-project/vllm#47311](#); on older builds pass `--tool-call-parser glm47` (the GLM 4.7 parser) instead of `poolside_v1`.

The FP8-quantized KV cache requires vLLM `>= 0.22.0`. Earlier versions produce scrambled output on non-Hopper GPUs because of a per-layer attention-head count bug, fixed in [vllm#42650](#). On older vLLM, disable the FP8 KV cache by adding `--kv-cache-dtype-skip-layers $(seq 0 39)`.

SGLang

The full SGLang recipe is on the main [Laguna XS 2.1 model card](#) and on the [SGLang cookbook](#). SGLang reads the checkpoint `quantization_config`, so you can use the same launch command after replacing the model ID

Transformers

The full Transformers recipe is on the main [Laguna XS 2.1 model card](#). Substitute poolside/Laguna-XS-2.1-INT4 for the model ID; quantization is detected automatically from `quantization_config`.

Ollama

Available on the [Ollama library](#).

macOS (Metal) users: Chat (`ollama run /api/chat`) works as expected on Linux/CUDA. On macOS/Metal it may currently return empty output; the root cause is not yet fully understood and we're investigating it with the Ollama team. On a Mac, use a Linux/CUDA host, or the `/api/generate` endpoint with `"raw": true`.

Controlling reasoning

Laguna XS 2.1-INT4 uses the same reasoning controls (interleaved thinking, preserved reasoning, and the `enable_thinking` flag) as the base model. See the [Controlling reasoning](#) section of the main Laguna XS 2.1 model card.

License

This model is licensed under the [OpenMDW-1.1 License](#).

Intended and Responsible Use

Laguna XS 2.1-INT4 is designed for software engineering and agentic coding use cases, and you are responsible for confirming that it is appropriate for your intended application. Laguna XS 2.1-INT4 is subject to the [OpenMDW-1.1 License](#), and should be



[Policy](#). We advise against circumventing Laguna XS

2.1-INT4 safety guardrails without implementing substantially equivalent mitigations appropriate for your use case.

Please report security vulnerabilities or safety concerns to security@poolside.ai.